

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

Chromosomes Structure

By

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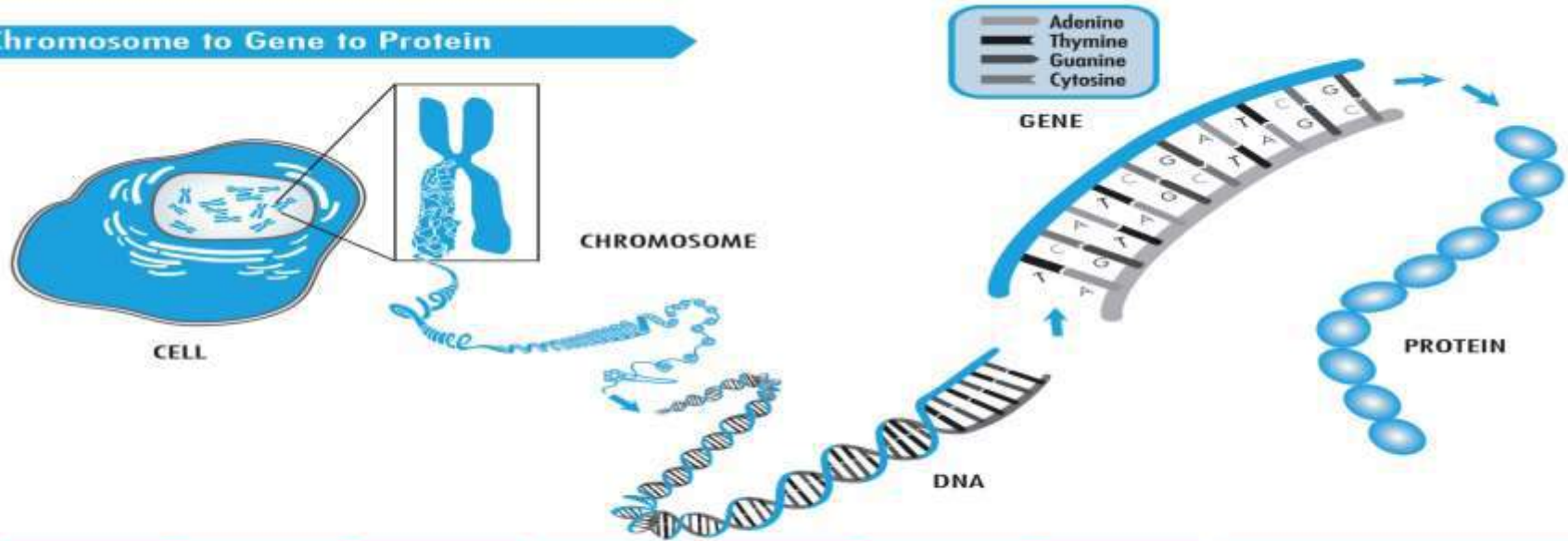
KAU

Objectives

- Differentiate between chromatin and chromosomes.
- Explain the significance and use of a karyotype.
- Describe the role of the centromere in connecting sister chromatids.
- Analyze how chromosomal structure supports proper cell division.

Chromosome to Gene to Protein

Chromosome to Gene to Protein



Chromosomes are like chapters in a book; one set is from the mother, one set is from the father.

Genes are like sentences in a chapter.

Chromosomes

Definition

- **Chromosomes** are thread like structures in the nucleus of cells that contain genetic material or DNA.
- **Chromosomes** are thread like structures that hold our genes.

✧ **Basic Definitions** ✧

Gamete

- Reproductive cells (i.e. Sperms & eggs).

Somatic cells

- All body cells except reproductive cells

Homologous chromosomes

- **Chromosomes of the same size and shape which carry the same type of genes.**

Chromatid

- **One of two duplicated chromosomes connected at the centromere**

Centromere

- **Region of chromosome where sister chromatid are hold together and microtubules attach during mitosis and meiosis**

Introduction

- Although the nucleus of a human cell is only about 5–8 μm long, it holds all the genetic material necessary to direct all the functions in the body.
- The genetic material within the nucleus is arranged into **chromosomes**, structures that assist in the transmission of genetic information from one generation to the next.

Introduction

- The instructions in each chromosome are contained within genes, which in turn are composed of DNA.
- Chromosomes also contain proteins (histone and non histone) that assist in the organizational structure of the chromosome.
- Collectively, the DNA and proteins are called **chromatin**.

Chromosomes

- Each species has a characteristic chromosome complement (karyotype) in terms of the number and the morphology of the chromosomes that make up its genome.
- In human there are 46 chromosomes (23 pairs)

Types of Chromosomes

- **Autosomes**

- Chromosomes other than the sex chromosomes
- In humans, there are **44** autosomes (22 pairs)

- **Sex chromosomes**

- The **X and Y** chromosomes are involved in sex determination.
- These have different sizes and shapes

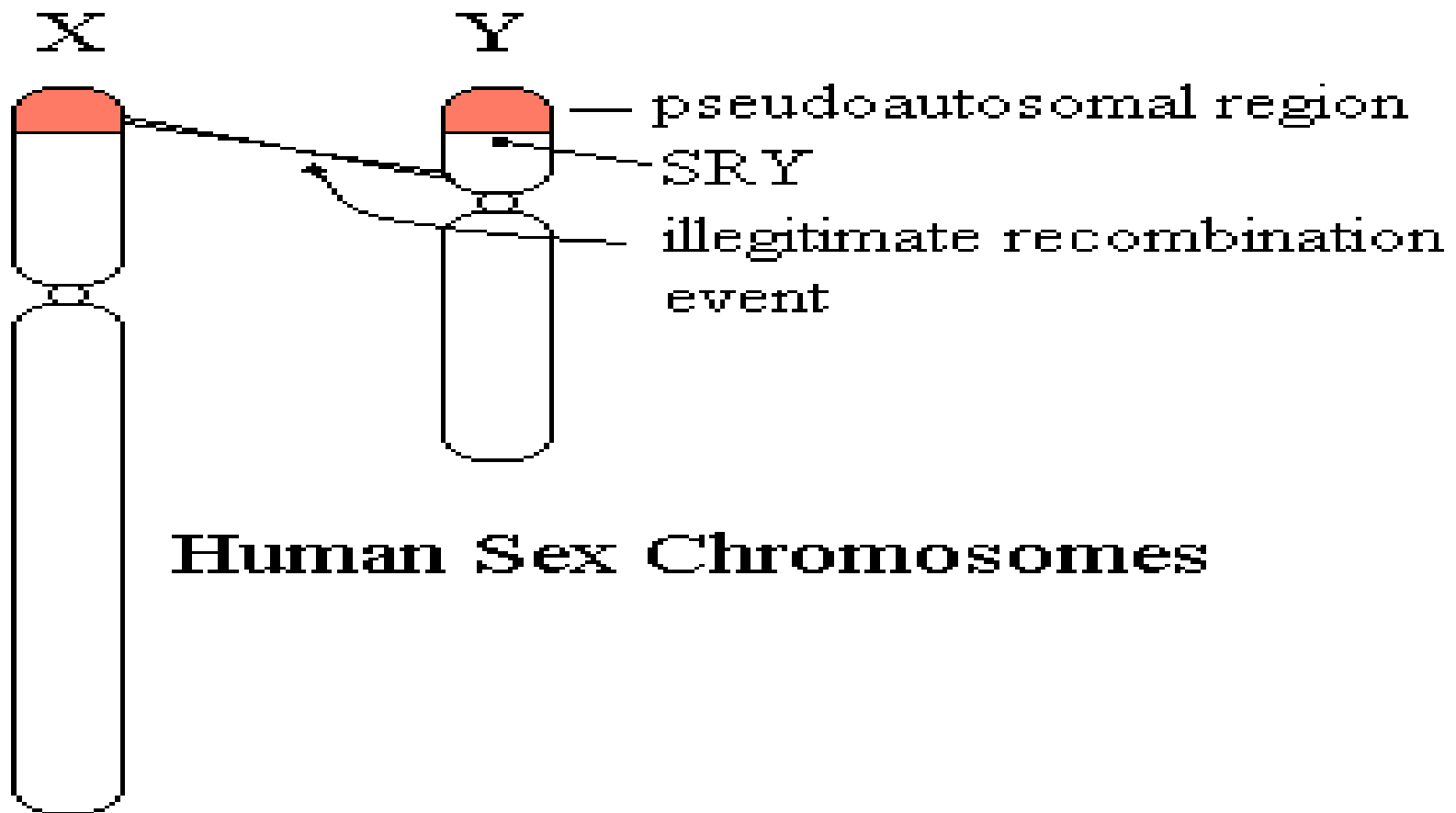
Human autosomes

- Each autosome, is numbered from 1 to 22 in order of **decreasing length**.
- There are two identical chromosomes for these chromosomes 1-22. These chromosomes are said to be **homologous pairs**. In other words, they have identical size, structure, and a similar nucleotide sequence.

Sex chromosomes

- The sex chromosomes X and Y are not homologues.
The X chromosome is longer than the Y chromosome.
- Males have the X and Y sex chromosomes, and females have two X chromosomes.
- The presence of the SRY gene on the Y chromosome causes testes to develop in males

Sex chromosomes



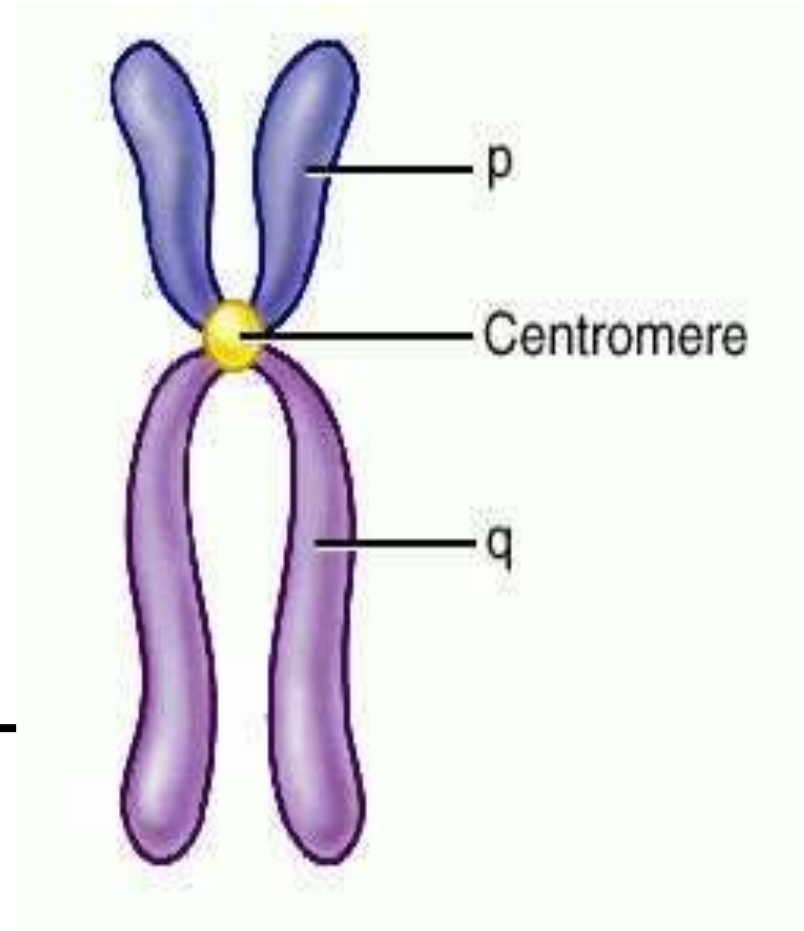
Chromosomal Morphology:

Each chromosome has a
constriction: **a centromere**

Short arm: **p arm**

Long arm: **q arm**

Telomere ———



Chromosome morphology

- Each chromosome can be seen to consist of two identical strands known as **chromatids**, or **sister chromatids**, which are the result of DNA replication taken place during the S (synthesis) phase of the cell cycle
- **Sister chromatids** contain the same genes.
- **Genes** are the units of heredity that control traits, such as eye color and metabolism.

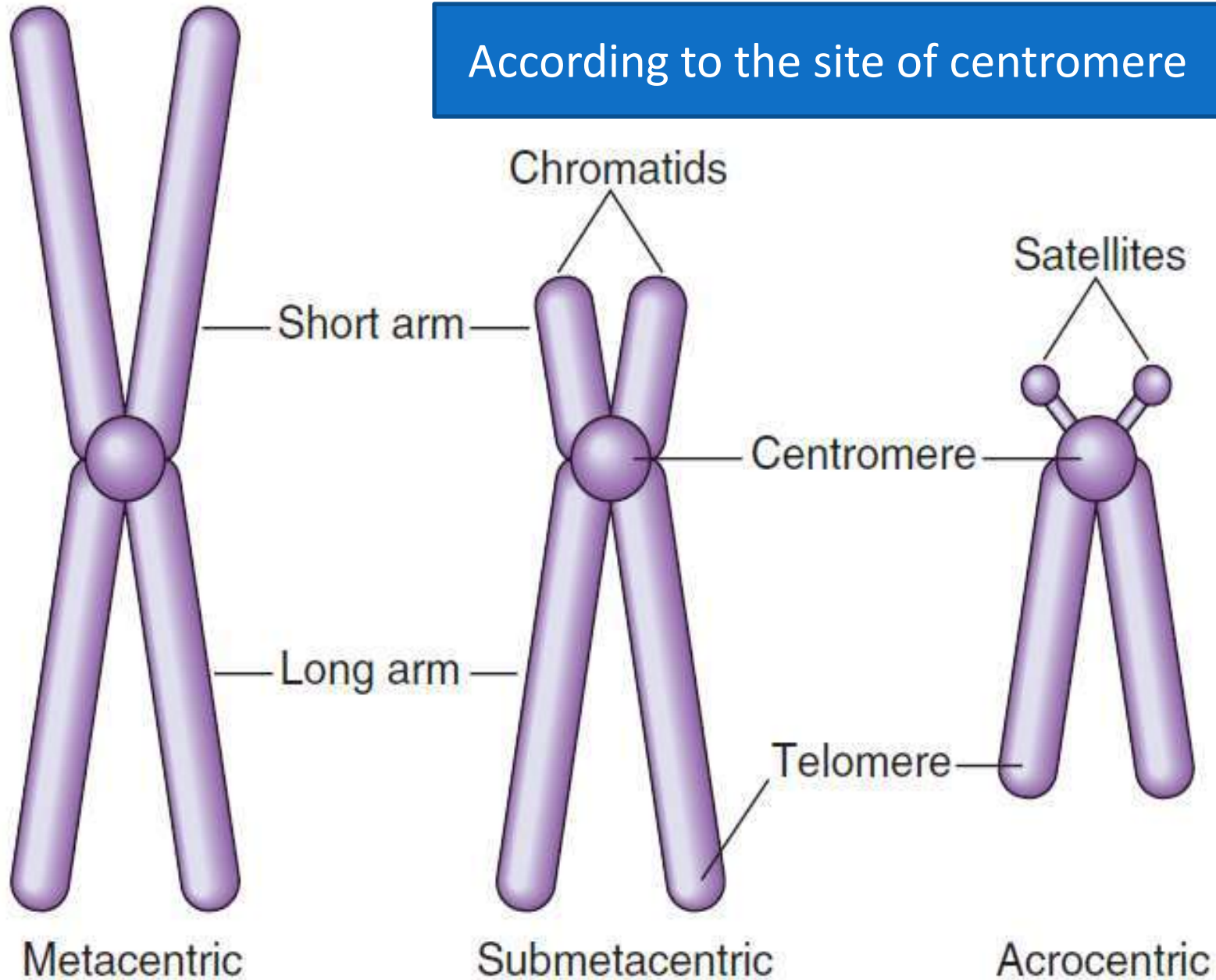
Chromosome morphology

- Humans have about **20,000 different genes**, and collectively they contain the instructions for how our bodies conduct the characteristics of life.
- These sister chromatids can be seen to be joined at a primary constriction known as the **centromere**.

Centromere

- Centromeres consist of several hundred kilobases of repetitive DNA (alpha satellite DNA) that are packaged into heterochromatin.
- The centromere serves as the site for kinetochore assembly and therefore it is essential during the alignment of the chromosomes at the metaphase plate and the subsequent segregation of chromosomes during cellular division.

According to the site of centromere



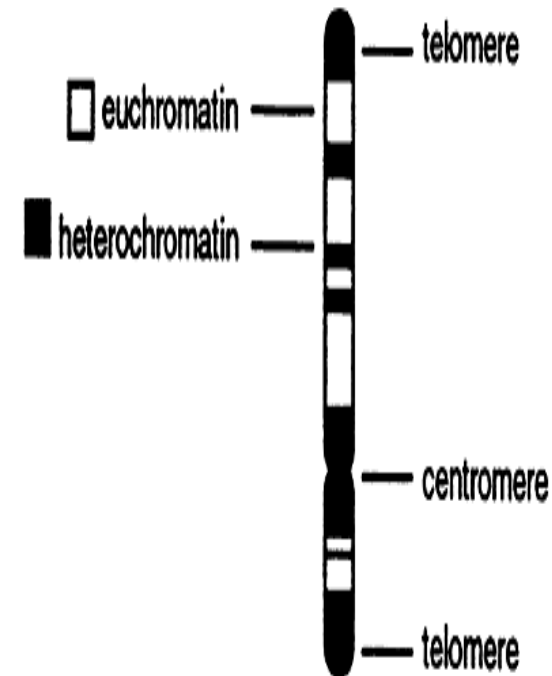
Telomere.

- The tip of each chromosome arm is known as the telomere.
- Telomeres play a crucial role in sealing the ends of chromosomes and maintaining their structural integrity.
- They consist of many tandem repeats of a TTAGGG sequence.

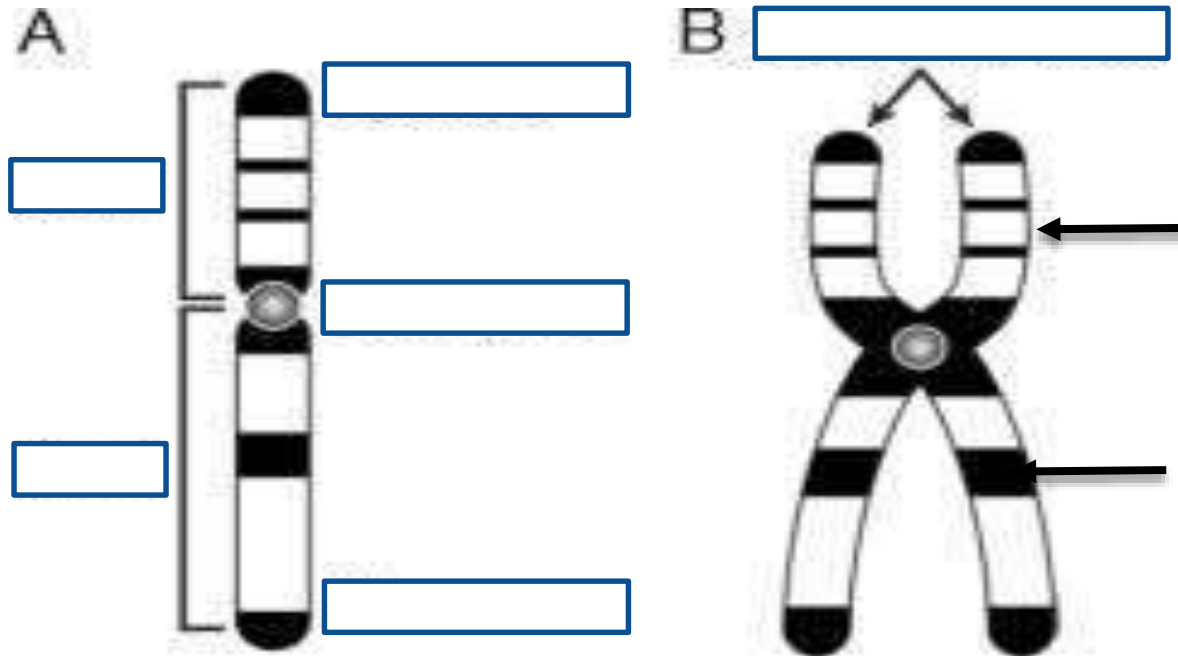
Structure of human chromosomes

Chromosomes have two structurally and functionally distinguishable sites:

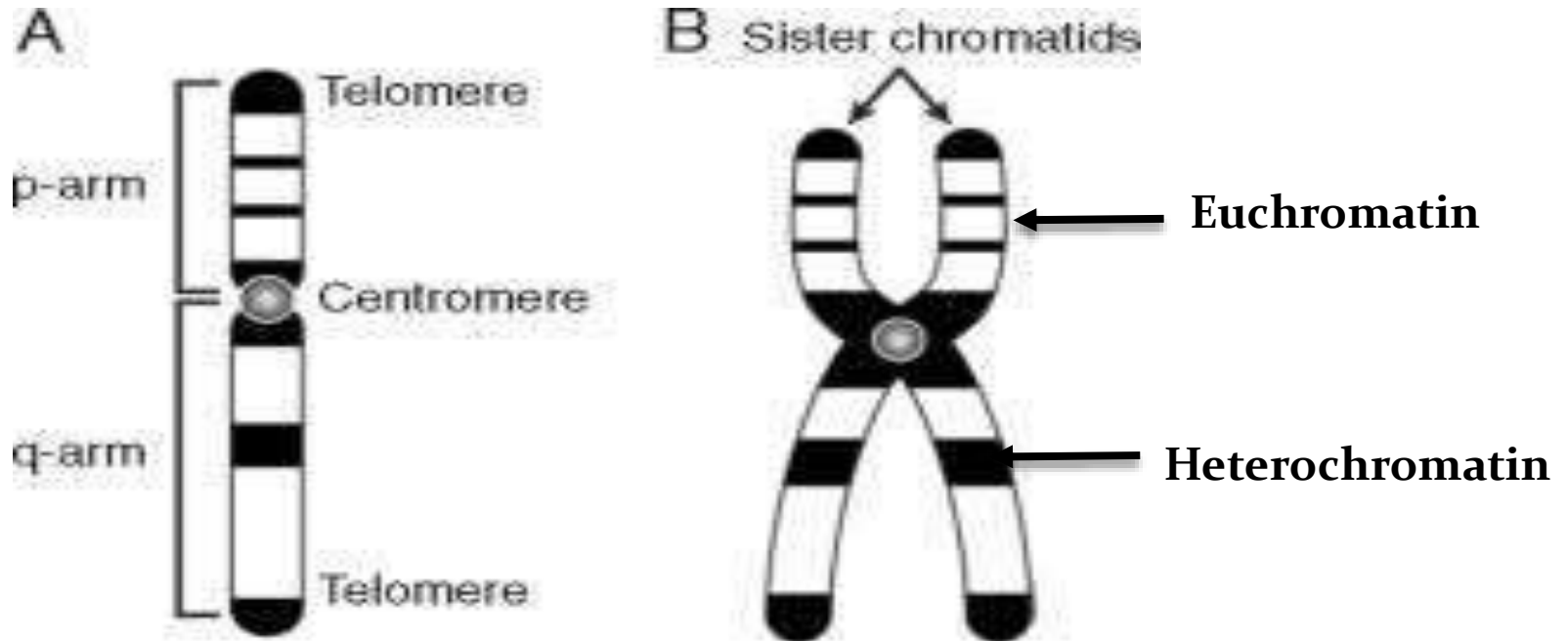
- **Heterochromatin** is highly condensed, gene poor and transcriptionally silent
- **Euchromatin** is less condensed, gene rich, and more easily transcribed.



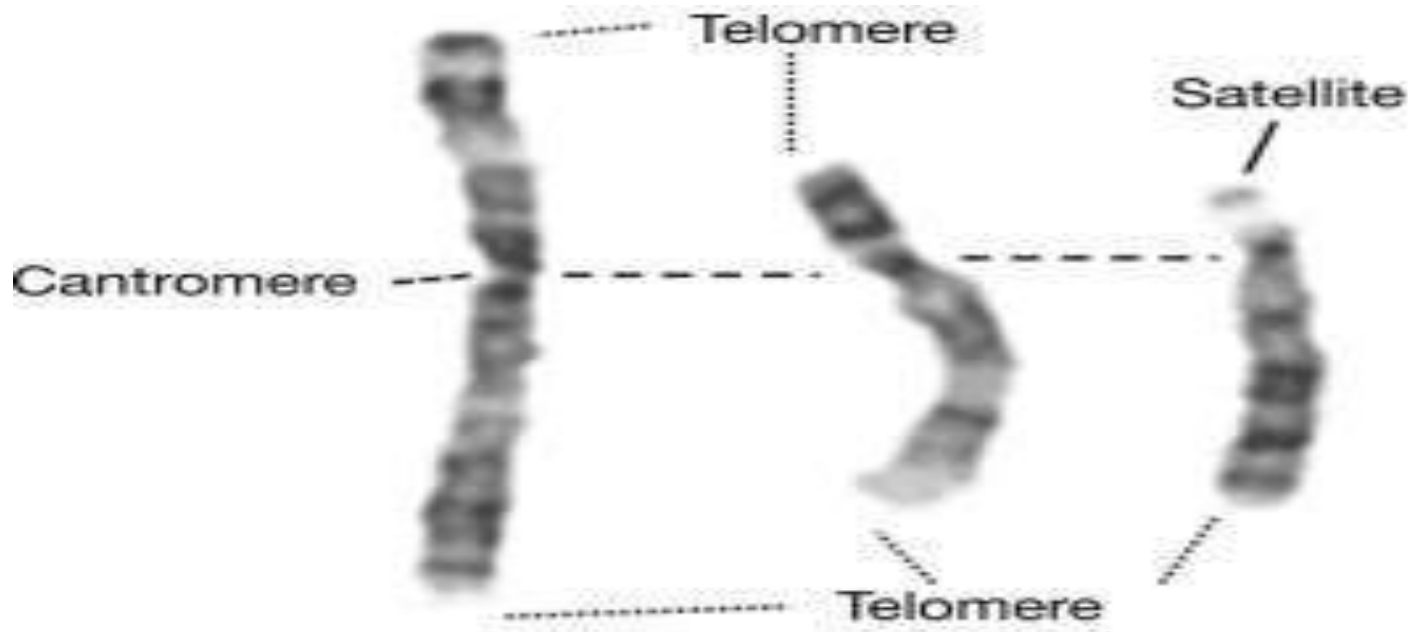
Recognize the basic structure of a chromosome under the microscope



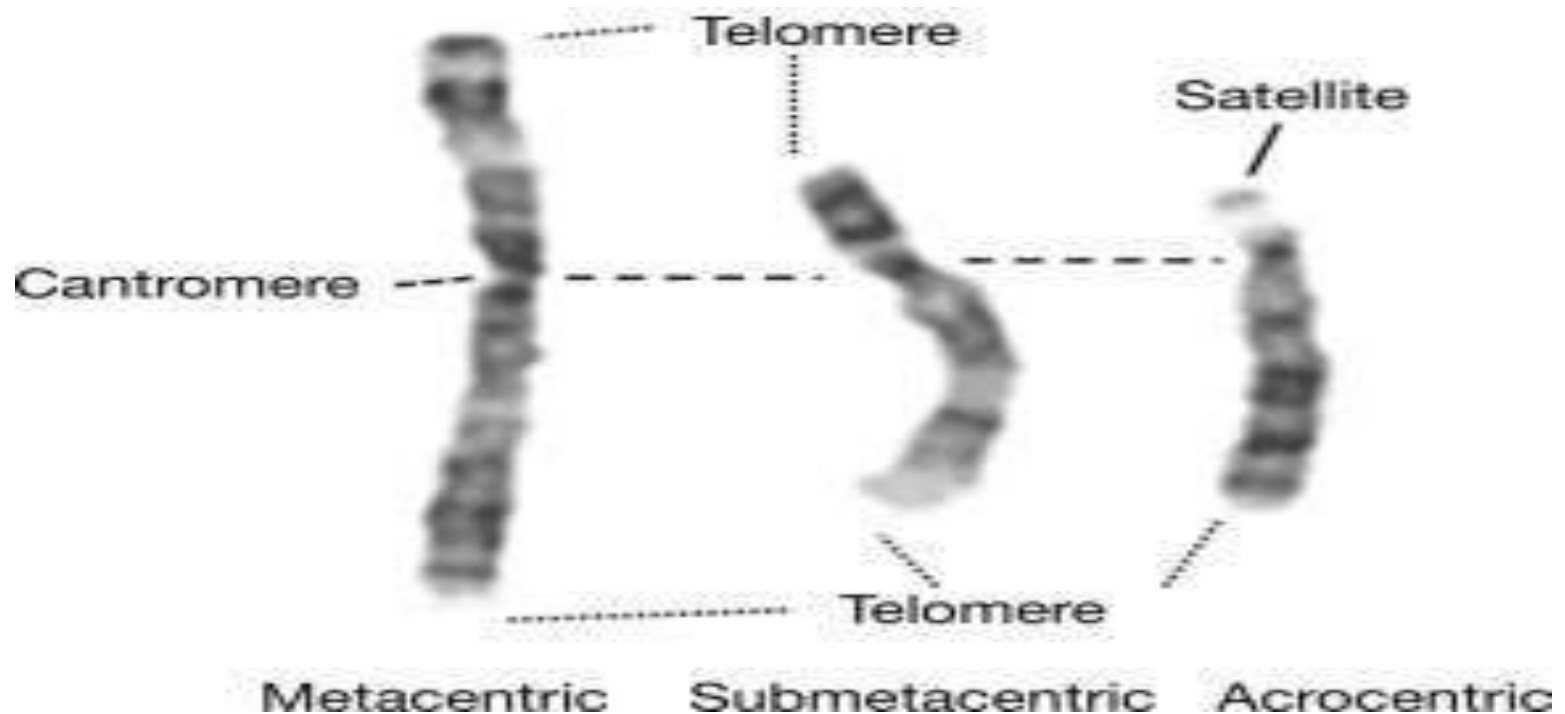
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Classify chromosomes as metacentric, sub metacentric, or acrocentric based on centromere position.



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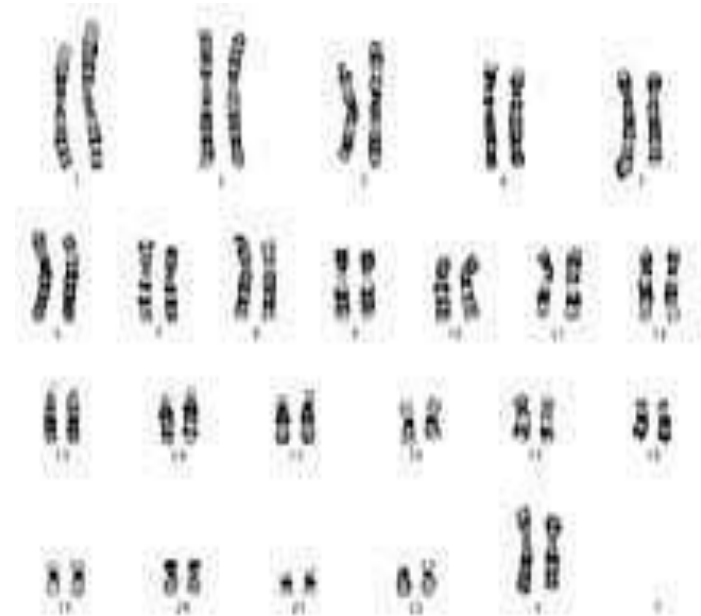
- Metacentric chromosomes: **1, 3, 16 , 19, 20**
- Sub metacentric chromosomes: **2,4,5, 6 - 12, 17,18 and X**
- Acrocentric chromosomes: **13,14,15,21,22 and Y**

Obtaining Cells for Chromosome Studies

- **Any nucleated cells can be used to make karyotype**
 - **Lymphocytes, skin cells, cells from biopsies, tumor cells**
- **Sampling cells before birth (fetal cells)**
 - **Amniocentesis**
 - **Chorionic villus sampling (CVS)**

Chromosomal Identification

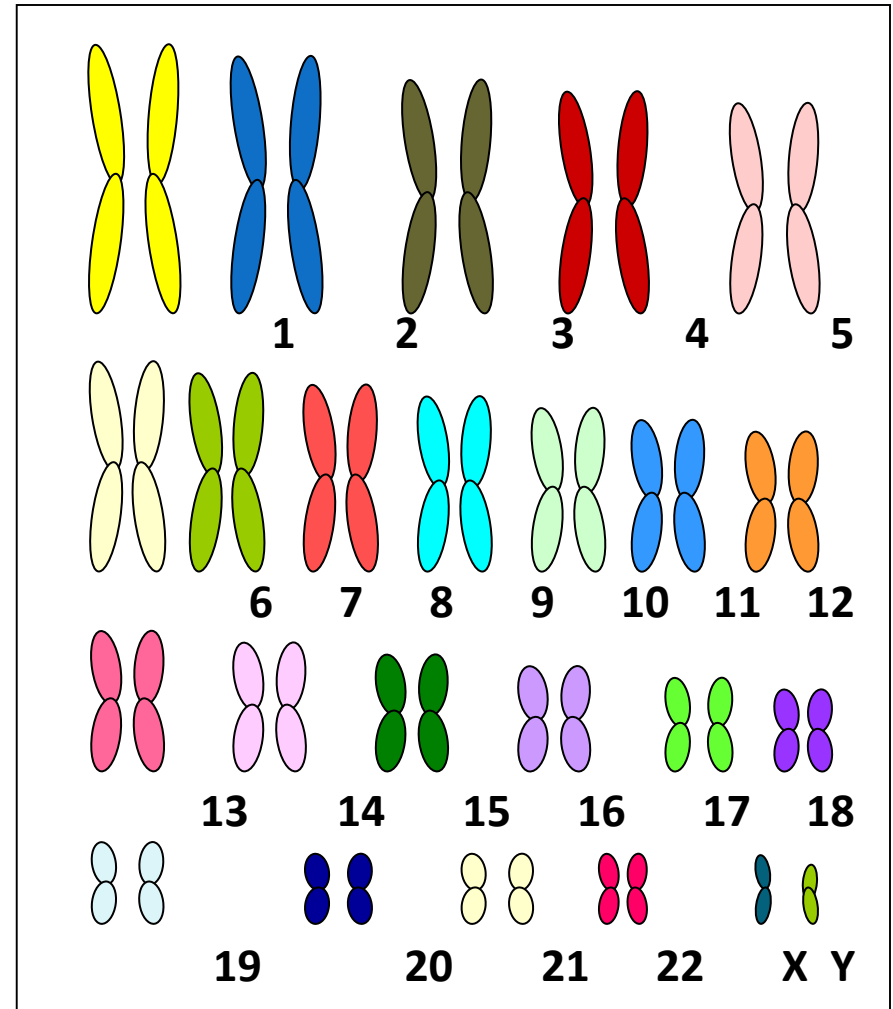
- Chromosomes can be identified by:
- Their size
- Their shapes (the position of the centromere)
- Banding pattern produced by specific stain (Giemsa stain)
- Chromosomes are analyzed by or organizing them into a karyogram.



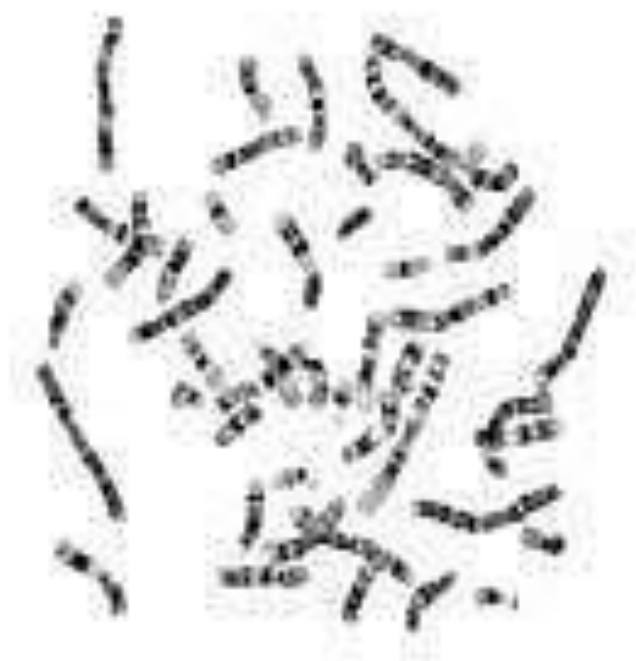
A Karyogram

Definition

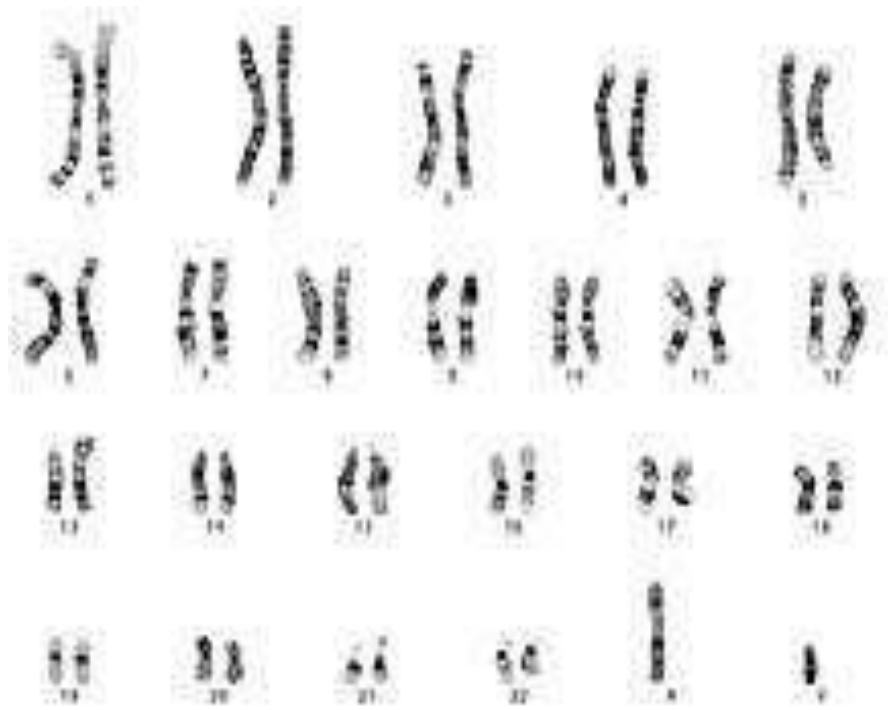
Photographic arrangement of a complete set of chromosomes of a cell or organism



Metaphase Chromosomes Arranged Into a Karyogram



metaphase



Karyogram

Banding techniques

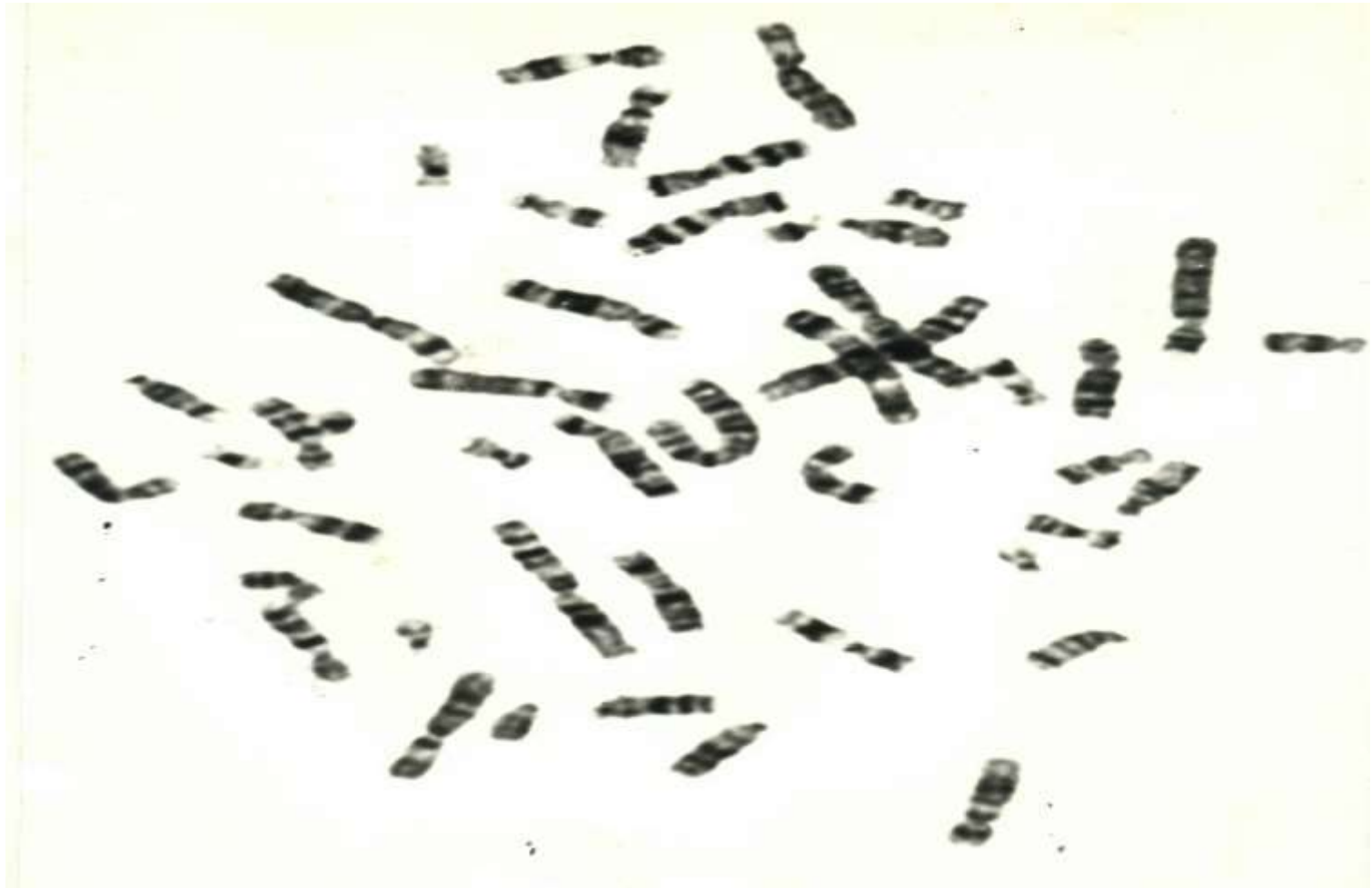
Banding procedures for karyotypes produce patterns of bands that are unique for each chromosome. Comparison of these bands can help to match homologous pairs in a karyotype.

Banding techniques:

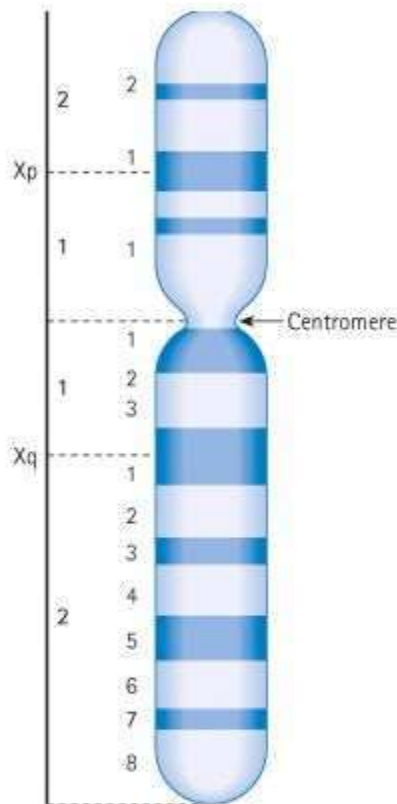
G-banding

- The technique most widely used in clinical cytogenetics laboratories. The chromosomes are treated first with **trypsin** to digest the chromosomal proteins and then with **Giemsa stain**.

G- banding metaphase



Nomenclature



An X chromosome showing the short and long arms each subdivided into **regions** & **bands**

Chromosome Labeling

Chromosome is identified with a number ranging 1-22, or X and Y



Each chromosome is divided into 2 arms (p,q)

Each arm divided into regions and identified by a number



Each region divided into bands identified with a number

Example - 1q24

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Each region divided into bands identified with a number

Example - 1q24

- The first chromosome, long arm, second region of the chromosome, the fourth band of that region.

How to write a cytogenetic karyotype

1. Total number of chromosomes
2. Sex chromosomes (XX or XY)
3. The abnormality
4. The chromosome number
5. Either short arm (p) or long arm (q)
6. Region
7. Band
8. Sub-band

e.g: 46,XY,del(15)(q13)

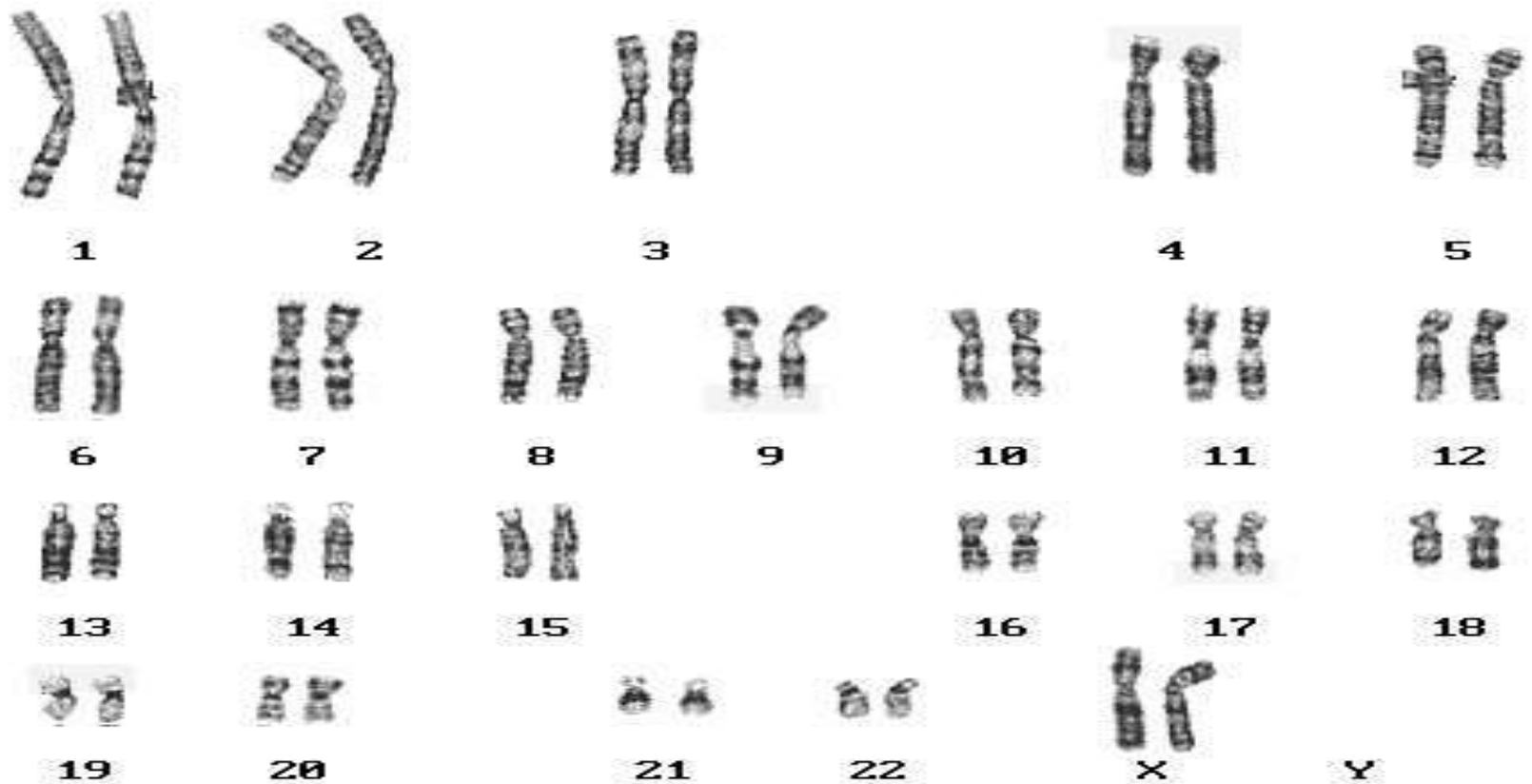
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(male karyotype showing terminal deletion of chromosome number 15 long arm region 1 band 3)

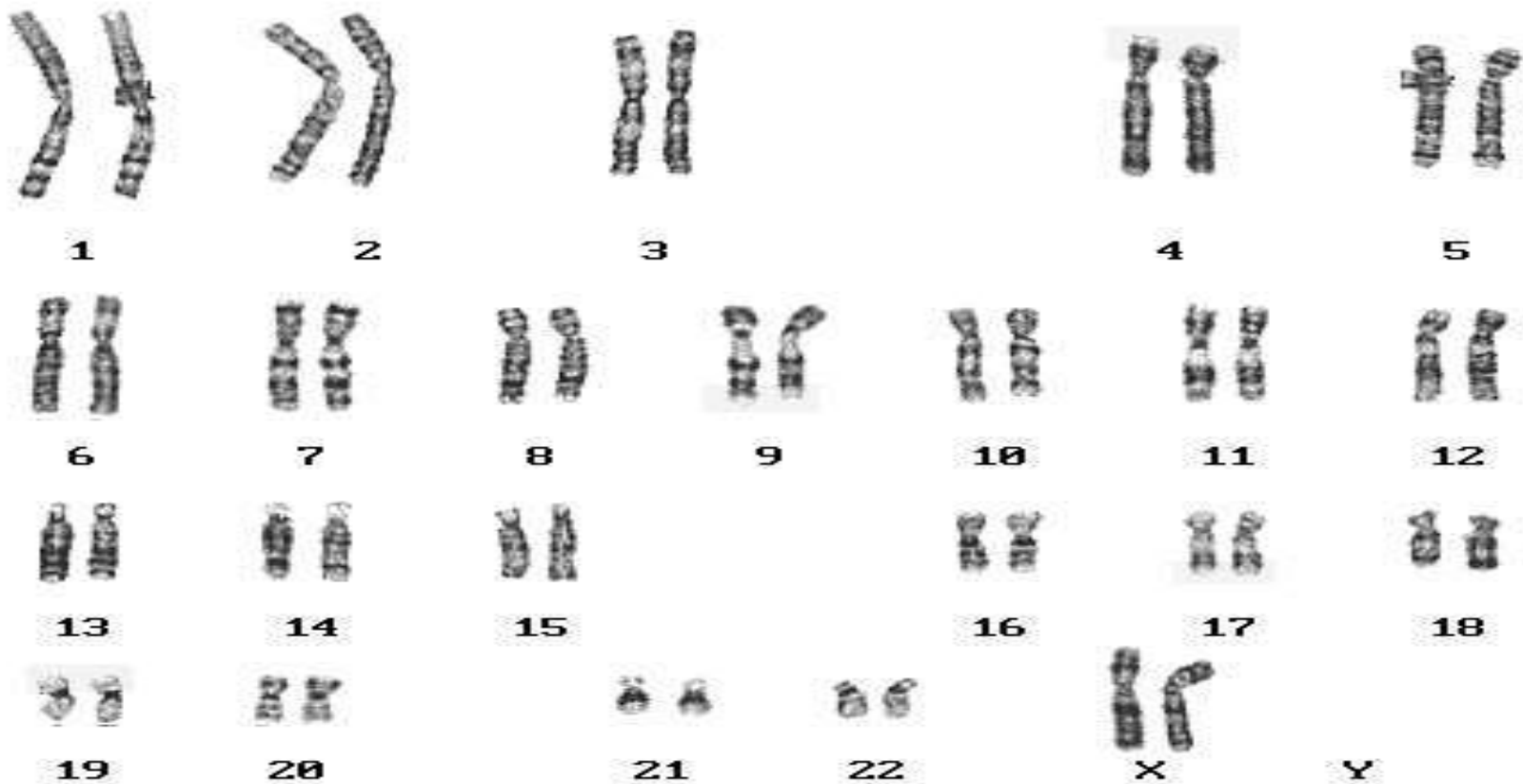
What is the nomenclature of the following Karyogram ?



Karyotype: 46, XX

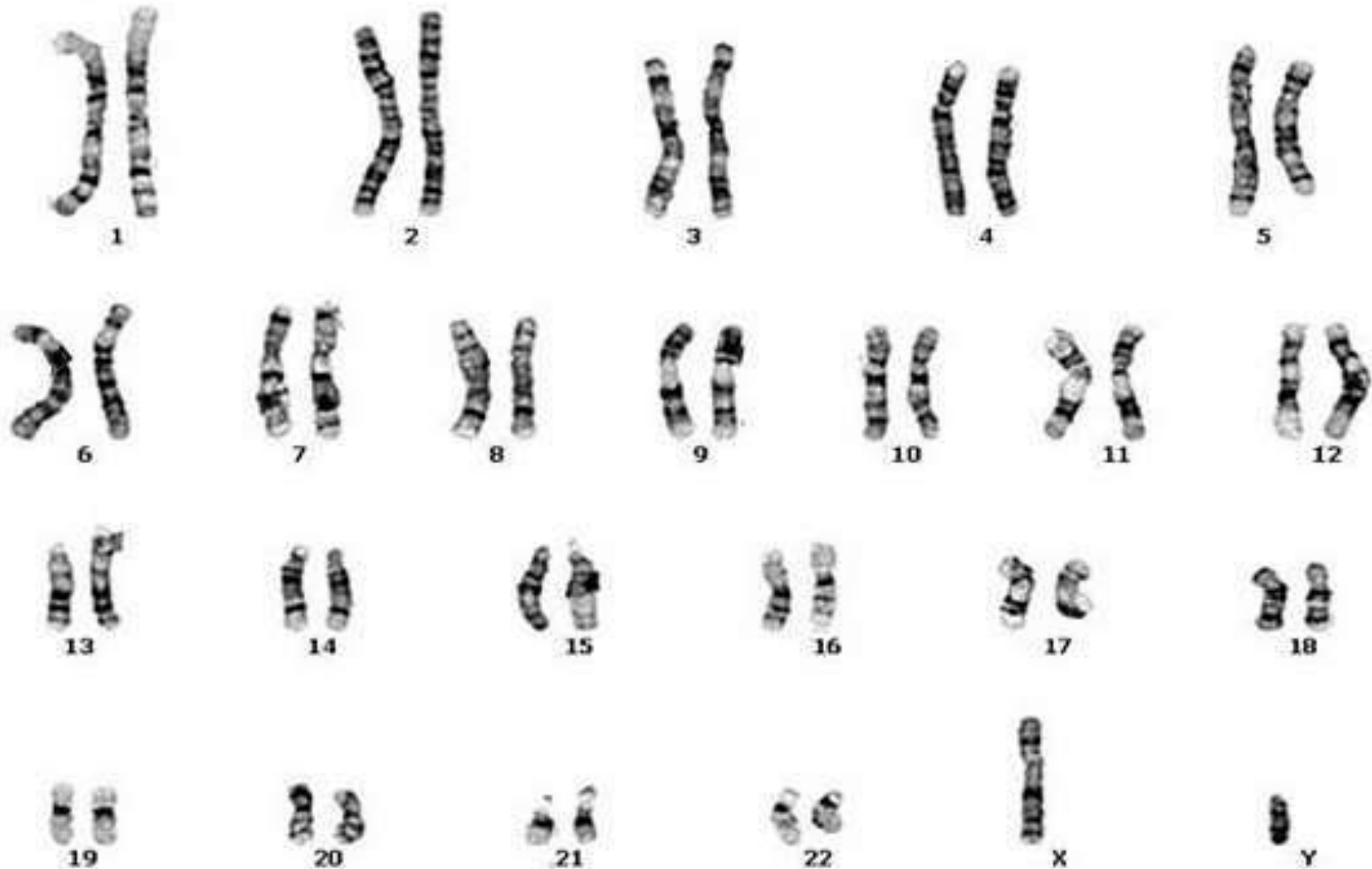
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46,XX



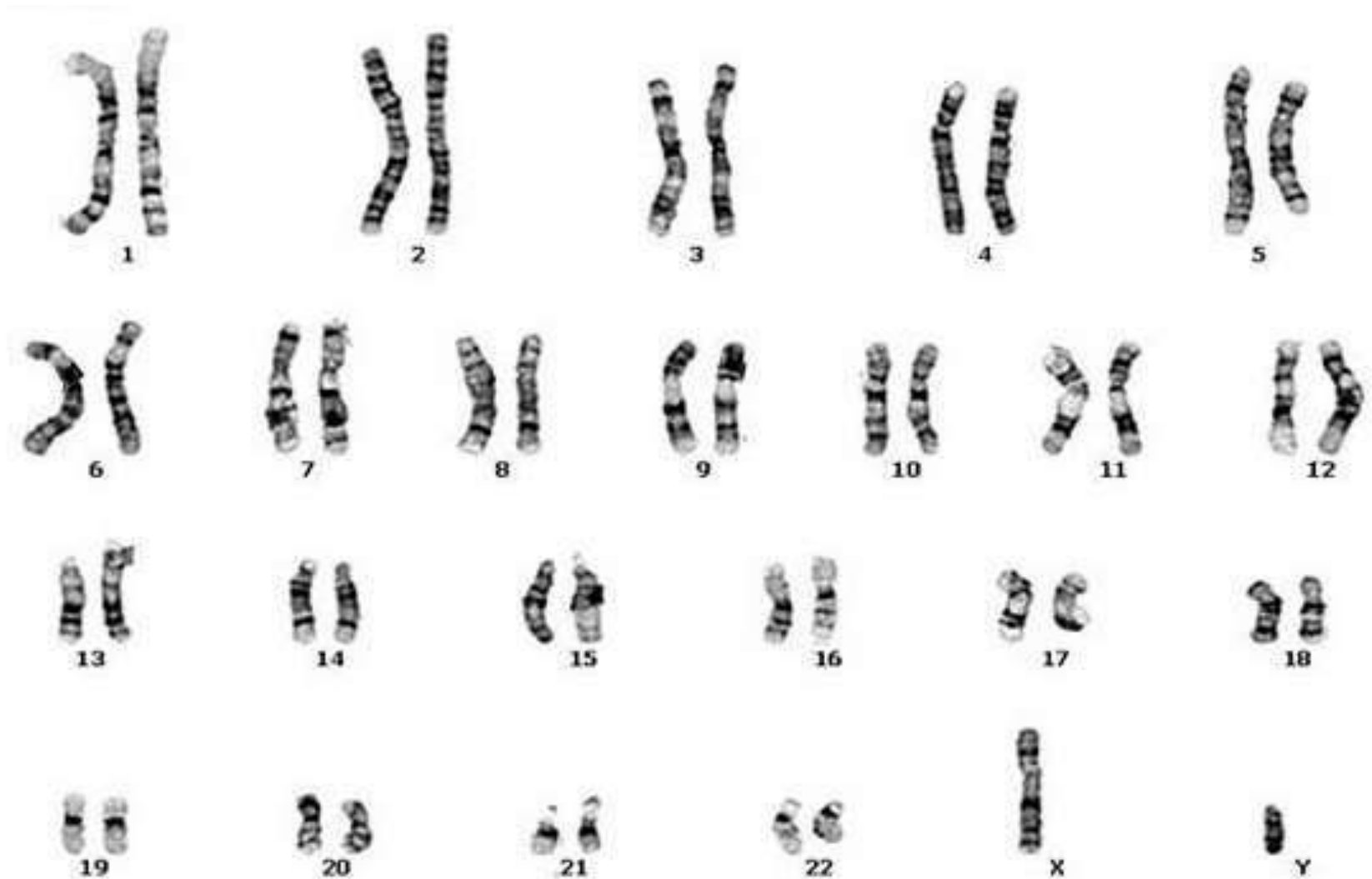
Karyotype: 46,XX

What is the nomenclature of the following Karyogram ?



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46,XY



- ***Chromosomal analysis is indicated under the following circumstances***

- ❖ Presence of a typical defined chromosomal disorder
e.g. Down syndrome.
- ❖ Presence of **four features** - mental retardation, physical retardation, malformations and dysmorphogenesis in a child.
- ❖ Features of two or more syndromes in one patient to exclude a contiguous gene syndrome.

● Chromosomal analysis is indicated under the following circumstances

- ❖ Malformations known to have a high association with a chromosomal disorder e.g. holoprosencephaly
- ❖ Child with non- specific dysmorphism without a specific diagnosis.

Indications for Chromosome analysis

- ❖ **Pregnancy in a woman of advanced age.**
- ❖ **Family history of a known or suspected chromosome abnormality**
- ❖ **Stillbirth and neonatal death.**
- ❖ **Neoplasia**
- ❖ **Fertility problems.**
- ❖ **amenorrhea**
- ❖ **infertility**
- ❖ **recurrent miscarriage.**

Obtaining Cells for Chromosome Studies

- **At which phase of the cell cycle chromosomal study can be done?**

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At Metaphase

What are chromosomes, which have identical size, structure, and a similar nucleotide sequence?

- Homologous chromosomes
- Sex Chromosomes
- Human autosomes
- Non-homologous chromosomes

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Which of the following is not used to identify chromosomes during chromosomal analysis?

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Which of the following chromosomal region is responsible for spindle's microtubules attachment during mitosis and meiosis?

- Telomeres
- Short arm (p)
- long arm (q)
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Which of the following is the number of autosomes in the human reproductive cells (sperm and ova)?

- 22 autosomes
- 46 autosomes
- 23 autosomes
- 44 autosomes



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Reference

- ❑ **[https:// images.google.com](https://images.google.com)**
- ❑ **Google Images. The most comprehensive image search on the web.**
- ❑ **Emery's Elements of Medical Genetics 15th edition 2017 by
Turnpenny and Ellard**

Thank You!

